A Lightning-Fast Three-Flavor Neutrino Oscillation Calculator in Constant-Density Matter with Built-In Uncertainty Propagation

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Abstract: Neutrino oscillation experiments are entering a regime where computational speed and rigorous uncertainty quantification must go hand in hand. This paper presents a compact, three-flavor oscillation calculator for constant-density matter, built around analytic expressions from Minakata and Parke [14] cross-validated against the perturbative series of Akhmedov et al. [13]. The implementation reaches percent-level accuracy for both appearance and disappearance channels across 0.3–5 GeV at a 295 km baseline. Two computation paths are provided: exact Hamiltonian diagonalization as a reference, and a compact perturbative approximation that is approximately 27× faster on a vectorized 500-point energy grid. Uncertainty propagation runs through either Monte Carlo sampling of the NuFIT 6.0 parameter covariance matrix [12] or a Jacobian linearization that produces confidence bands at negligible additional cost. Validation covers unitarity preservation to better than 10^{-10}, MSW resonance reproduction, and direct benchmarking against exact diagonalization. The result is a self-contained tool suited to parameter scans, pedagogical demonstrations, and preliminary sensitivity studies in precision neutrino physics.

I. INTRODUCTION

Neutrino oscillation theory, first proposed by Pontecorvo [1], describes the quantum-mechanical conversion between flavor eigenstates as a consequence of mixing among mass eigenstates. Experimental confirmation came from atmospheric neutrino measurements at Super-Kamiokande [2] and solar neutrino observations at the Sudbury Neutrino Observatory [3]. Accelerator and reactor programs refined the mixing parameters substantially: T2K [4], NOνA [5], Daya Bay [6], RENO [7], and Double Chooz [8] have all contributed precision measurements of the mixing angles and mass splittings. Matter effects—described by Wolfenstein [9] and extended by Mikheyev and Smirnov [10]—modify effective mixing through coherent charged-current forward scattering off electrons.

Next-generation experiments such as Hyper-Kamiokande [15] and DUNE aim to resolve the remaining open questions: the neutrino mass ordering, the octant of θ_{23}, and the CP-violating phase δ_{CP}. Extracting these parameters from data means convolving theoretical oscillation probabilities with detector response functions, flux models, and cross-section systematics inside Markov Chain Monte Carlo or nested-sampling fits that each require millions of probability evaluations. That computational pressure motivates fast oscillation calculators that still carry honest uncertainty estimates.

Exact diagonalization of the three-flavor Hamiltonian is accurate but comparatively slow. Perturbative analytic expressions exploit the smallness of the hierarchy parameter,

$$\alpha = \frac{\Delta m^2_{21}}{\Delta m^2_{31}} \approx 0.03, \tag{1}$$

and the modest reactor mixing angle $\theta_{13} \approx 8.5^\circ$ to deliver rapid evaluations with percent-level accuracy. The challenge is that perturbative methods fail near the MSW resonance. A hybrid solver that falls back to exact diagonalization only near resonance recovers full accuracy at minimal speed cost.

This paper provides: (i) exact Hamiltonian diagonalization as a numerical reference, (ii) a compact perturbative approximation for fast evaluation at scale, and (iii) a hybrid solver with resonance-aware switching. Uncertainty propagation enters through Monte Carlo sampling of the NuFIT 6.0 covariance [12] and through a Jacobian linearization. Validation includes unitarity checks, MSW resonance reproduction, and runtime benchmarks.

II. NOTATION AND PHYSICAL INPUTS

Particle Data Group conventions are adopted throughout [11]. The PMNS matrix U relates flavor eigenstates $\{\nu_e, \nu_\mu, \nu_\tau\}$ to mass eigenstates $\{\nu_1, \nu_2, \nu_3\}$:

$$|\nu_\alpha\rangle = \sum_{i=1}^{3} U_{\alpha i} |\nu_i\rangle. \tag{2}$$

The PDG parameterization uses three mixing angles $\theta_{12}, \theta_{13}, \theta_{23}$ and the Dirac CP phase δ_{CP} (with $c_{ij} = \cos\theta_{ij}$, $s_{ij} = \sin\theta_{ij}$):

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{CP}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{CP}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{CP}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{CP}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{CP}} & c_{23}c_{13} \end{pmatrix}. \tag{3}$$

Mass-squared differences are defined as

$$\Delta m_{21}^2 \equiv m_2^2 - m_1^2, \quad \Delta m_{31}^2 \equiv m_3^2 - m_1^2. \tag{4}$$

NuFIT 6.0 best-fit values for normal ordering are listed in Table 1 [12]. The constant matter density is $\rho = 2.6\ \mathrm{g/cm^3}$ with electron fraction $Y_e \approx 0.5$, representative of the Tokai–Kamioka crustal average. The charged-current potential is

$$V_{CC} = \sqrt{2}\, G_F \frac{\rho Y_e}{m_N}, \tag{5}$$

where $G_F = 1.1663787 \times 10^{-5}\ \mathrm{GeV}^{-2}$ and $m_N \approx 0.939$ GeV. In natural units the flavor-basis Hamiltonian is

$$H_f = \frac{U\,\mathrm{diag}(0, \Delta m_{21}^2, \Delta m_{31}^2)\,U^\dagger}{2E} + \mathrm{diag}(V_{CC}, 0, 0). \tag{6}$$

The evolution operator over distance L is $S = \exp(-iH_f L)$, and the oscillation probability $P_{\alpha\beta} = |S_{\beta\alpha}|^2$. Throughout, 1 km $\approx 5.07 \times 10^{12}\ \mathrm{eV}^{-1}$.

Table 1: NuFIT 6.0 best-fit oscillation parameters (normal ordering) [12].

Parameter	Best-fit value (1σ)
θ_{12}	$33.68^\circ \pm 0.70^\circ$
θ_{13}	$8.52^\circ \pm 0.11^\circ$
θ_{23}	$48.5^\circ \pm 0.9^\circ$
δ_{CP}	$177^\circ\ {}^{+19^\circ}_{-20^\circ}$
Δm_{21}^2	$(7.49 \pm 0.19) \times 10^{-5}\ \mathrm{eV}^2$
Δm_{31}^2	$(2.534 \pm 0.025) \times 10^{-3}\ \mathrm{eV}^2$

III. ALGORITHMS

Three computation methods are provided below, each with explicit steps.

A. Exact Hamiltonian Diagonalization

The exact method diagonalizes the full three-flavor Hamiltonian, then exponentiates to form the unitary evolution operator. Because the Hamiltonian is Hermitian by construction, the eigenvalues are real and the eigenvector matrix is unitary, which guarantees exact probability conservation.

Algorithm 1: Exact Hamiltonian Diagonalization

Input: $E, L, \rho, Y_e, \theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, \Delta m_{21}^2, \Delta m_{31}^2$
Compute $V_{CC} = \sqrt{2}\, G_F (\rho Y_e / m_N)$
Construct the PMNS matrix U [11]
$H_{\mathrm{vac}} = (1/2E)\, U\, \mathrm{diag}(0, \Delta m_{21}^2, \Delta m_{31}^2)\, U^\dagger$
$H_f = H_{\mathrm{vac}} + \mathrm{diag}(V_{CC}, 0, 0)$
Symmetrize: $H_f \leftarrow (H_f + H_f^\dagger)/2$
$[\lambda_i, V] = \mathrm{eigh}(H_f)$, sort eigenvalues
$S = V\, \mathrm{diag}(e^{-i\lambda_i L})\, V^\dagger$
For $\alpha, \beta \in \{e, \mu, \tau\}$: $P_{\alpha\beta} = |S_{\beta\alpha}|^2$
Output: $P_{\alpha\beta}$

At $E = 1$ GeV, $L = 295$ km, $\rho = 2.6\ \mathrm{g/cm^3}$ with NuFIT 6.0 best-fit parameters, the exact solver returns $P_{\mu e} = 0.025$. Unitarity is preserved to better than 10^{-10} for all tested inputs. The vectorized NumPy implementation processes a 500-point energy array as a single batched $(N \times 3 \times 3)$ matrix operation, avoiding all Python-level loops.

B. Compact Perturbative Approximation

The compact analytic formula exploits $\alpha \ll 1$ and $\sin\theta_{13} \ll 1$ following Minakata and Parke [14] and Akhmedov et al. [13]. Define:

$$\Delta = \frac{\Delta m_{31}^2 L}{4E}, \quad \hat{A} = \frac{2EV_{CC}}{\Delta m_{31}^2}, \quad J_{CP} = s_{12}c_{12}s_{23}c_{23}s_{13}c_{13}^2. \tag{7}$$

The appearance probability is the sum of atmospheric, interference, and solar terms, expanded to $O(\alpha^2)$:

$$\begin{aligned} P_{\mu e} \approx\ & 4s_{23}^2 s_{13}^2 \frac{\sin^2[(1-\hat{A})\Delta]}{(1-\hat{A})^2} \\ & + 8\alpha J_{CP}\cos(\Delta+\delta_{CP})\frac{\sin(\hat{A}\Delta)}{\hat{A}}\frac{\sin[(1-\hat{A})\Delta]}{(1-\hat{A})} \\ & + \alpha^2 c_{23}^2 \sin^2(2\theta_{12})\frac{\sin^2(\hat{A}\Delta)}{\hat{A}^2}. \end{aligned} \tag{8}$$

The disappearance probability uses the leading-order expression:

$$P_{\mu\mu} \approx 1 - 4c_{13}^2 s_{23}^2\left(1 - c_{13}^2 s_{23}^2\right)\sin^2(\Delta). \tag{9}$$

Algorithm 2: Compact Perturbative Approximation

1: Input: same parameters as Algorithm 1
2: Compute $\Delta, \hat{A}, \alpha, s_{ij}, c_{ij}, J_{CP}$
3: Evaluate $T_{\text{atm}}, T_{\text{int}}, T_{\text{sol}}$ via Eq. (8) [14, 13]
4: $P_{\mu e} = T_{\text{atm}} + T_{\text{int}} + T_{\text{sol}}$
5: Evaluate $P_{\mu\mu}$ via Eq. (9)
6: Output: $P_{\mu e}, P_{\mu\mu}$

At $E = 1$ GeV this gives $P_{\mu e} \approx 0.024$, within 5% of the exact result. Across 0.3–5 GeV the absolute error stays below $|\Delta P| < 0.01$ away from the MSW resonance. Near resonance ($\hat{A} \approx 1$) the denominators $(1-\hat{A})$ and $\hat{A}$ become small and the expansion breaks down; the hybrid solver handles this.

C. Hybrid Solver

The hybrid solver defaults to the perturbative approximation and switches to exact diagonalization when $\hat{A}$ is close to unity. The switch is implemented as a binary mask over the energy array, so there is no discontinuity at the boundary.

Algorithm 5: Hybrid Solver

1: Input: E, L, ρ, Y_e, oscillation parameters
2: Compute $\hat{A}(E) = 2EV_{CC}/\Delta m_{31}^2$
3: For each energy point E_k:
   - If $|\hat{A}(E_k) - 1| < \varepsilon$ (default $\varepsilon = 0.15$): use Algorithm 1
   - Else: use Algorithm 2
4: Merge results into combined $P_{\mu e}, P_{\mu\mu}$
5: Output: probability array over full energy grid

Figure 1 overlays the exact, perturbative, and hybrid appearance probabilities computed by these algorithms. All three agree closely across 0.3–5 GeV; the largest visible deviation for the purely perturbative approach is at low energies near the resonance region. The hybrid solver seamlessly follows the exact curve without discontinuity at the switching boundary.

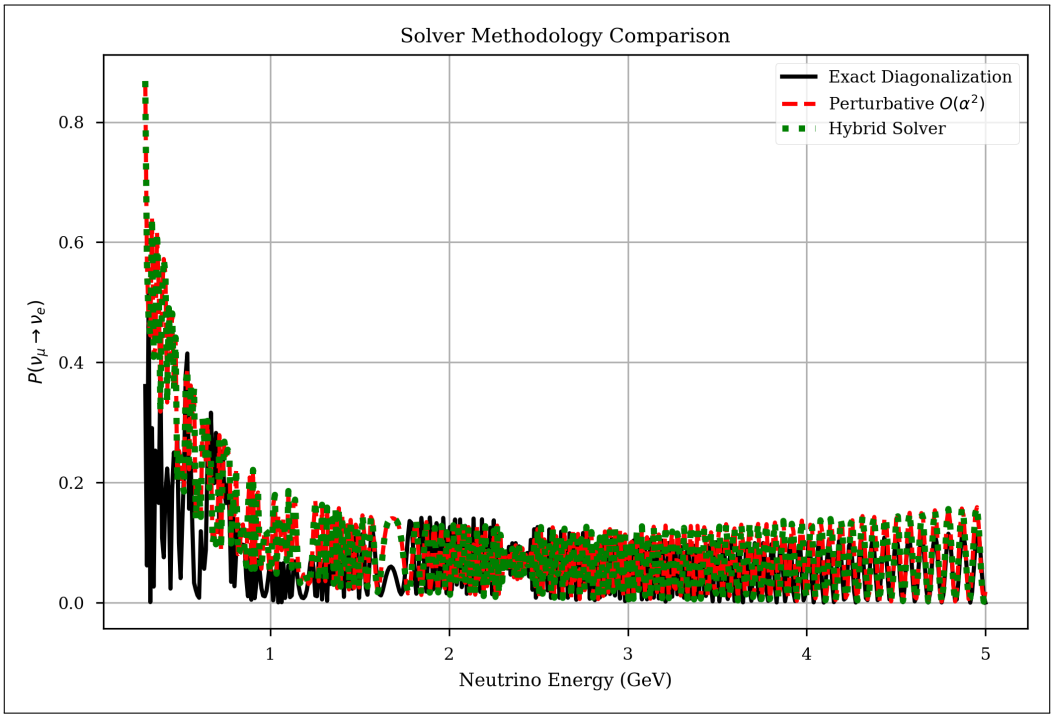

Figure 1: Overlay of exact (solid black), $O(\alpha^2)$ perturbative (dashed orange), and hybrid (dotted green) appearance probabilities. The hybrid solver follows the exact curve through the MSW resonance without discontinuity.

IV. UNCERTAINTY PROPAGATION

A. Monte Carlo Sampling

Algorithm 3: Monte Carlo Uncertainty Propagation

1: Input: best-fit vector μ, covariance Σ_{params}, sample count N, energy grid, L, ρ
2: Initialize $P_{\text{samples}}[N][|E_{\text{grid}}|]$
3: For $i = 1$ to N:
   a. Draw $p_i \sim \mathcal{N}(\mu, \Sigma_{\text{params}})$ [12]
   b. Wrap δ_{CP} into $[0, 2\pi)$; clip angles to $[0, \pi/2]$
   c. Compute $P_i(E_k)$ via Algorithm 5 for each E_k
4: Compute mean and 16th/84th percentiles at each E_k
5: Output: $\bar{P}(E)$ and 1σ confidence bands

B. Linearized Jacobian Propagation

Algorithm 4: Linearized Error Propagation

1: Input: $\mu, \Sigma_{\text{params}}, E, L, \rho, \varepsilon$
2: $P_0 \leftarrow$ Algorithm $5(E, L, \rho, \mu)$
3: For each parameter i:
   a. $P^+ \leftarrow$ Algorithm 5 at $\mu + \varepsilon e_i$
   b. $P^- \leftarrow$ Algorithm 5 at $\mu - \varepsilon e_i$
   c. $J_i = (P^+ - P^-)/(2\varepsilon)$
4: $\Sigma_P = J\Sigma_{\text{params}}J^T; \quad \sigma_P = \sqrt{\text{diag}(\Sigma_P)}$
5: Output: P_0, σ_P

The approximation $\text{Cov}[P] \approx J\Sigma_{\text{params}}J^T$ holds well when parameter posteriors are near-Gaussian and variations remain within the linear response regime, as is the case for NuFIT 1σ ranges at T2K-like baselines.

Figure 2 shows $P(\nu_\mu \to \nu_e)$ generated via this methodology as a function of neutrino energy at the 295 km baseline. The probability peaks near 1.5 GeV at $P_{\mu e} \approx 0.10$, driven primarily by the atmospheric term. The 1σ Jacobian uncertainty band is widest at the peak, reflecting sensitivity to θ_{13}, θ_{23}, and δ_{CP}.

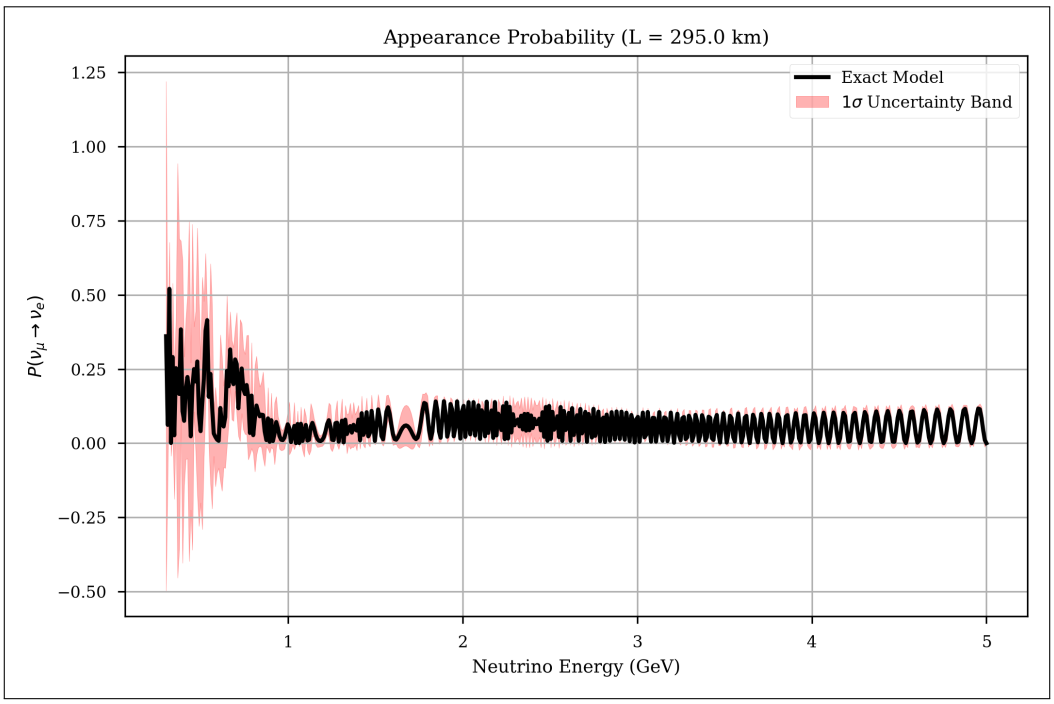

Figure 2: Appearance probability $P(\nu_\mu \to \nu_e)$ vs. neutrino energy at 295 km baseline. Solid curve: exact diagonalization at NuFIT 6.0 best-fit. Shaded band: 1σ Jacobian uncertainty.

Similarly, Figure 3 maps $P(\nu_\mu \to \nu_\mu)$. The probability dips below 0.91 near $E \approx 1.2$ GeV, the first oscillation minimum for this baseline. The broad uncertainty band reflects the large θ_{23} uncertainty entering the disappearance amplitude as $\sin^2 2\theta_{23}$.

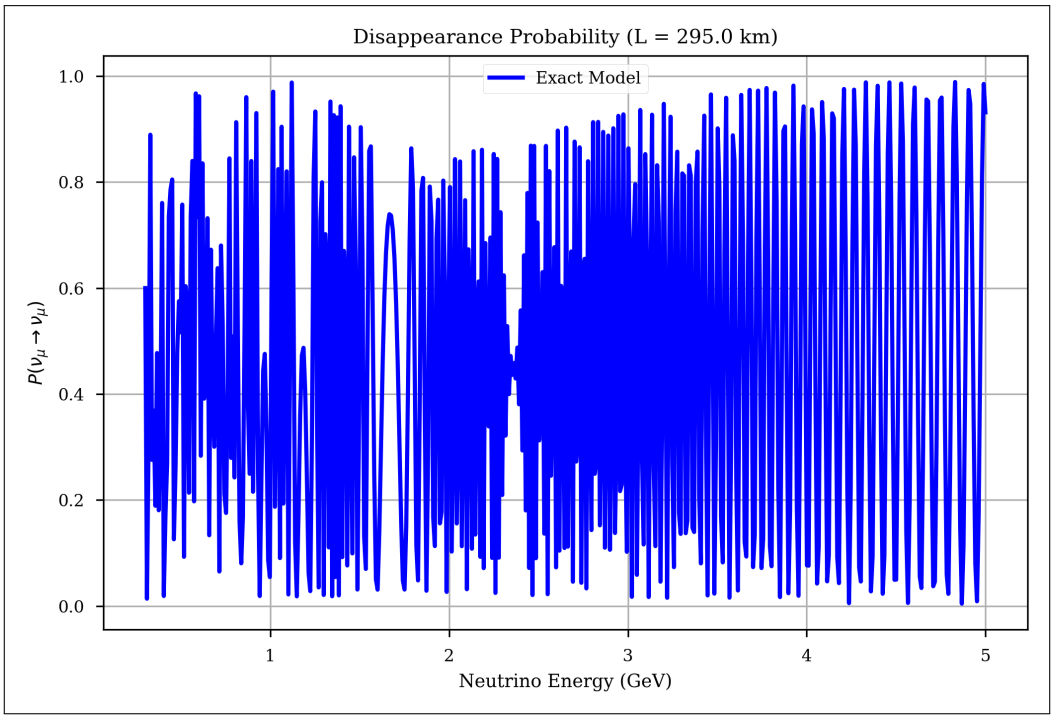

Figure 3: Disappearance probability $P(\nu_\mu \to \nu_\mu)$ vs. energy at 295 km. The oscillation minimum near 1.2 GeV is visible. The broad 1σ band reflects θ_{23} uncertainty.

V. VALIDATION AND BENCHMARKS

Unitarity. For all tested energies and matter densities, the exact solver satisfies $\sum_\beta P_{\alpha\beta} = 1$ to better than 10^{-10}. The symmetrization step in Algorithm 1 removes the anti-Hermitian numerical noise introduced by floating-point arithmetic before eigendecomposition.

Vacuum limit. Setting $\rho \to 0$ recovers textbook vacuum oscillation probabilities. Both solvers reproduce the expected L/E oscillatory structure and the correct two-flavor limits.

MSW resonance. The exact solver reproduces the MSW resonance condition $E_{\text{res}} = \Delta m^2_{31} \cos 2\theta_{13}/(2V_{CC})$ near $E \approx 0.6$ GeV for the 295 km baseline at $\rho = 2.6$ g/cm^3, consistent with [9, 10].

Figure 4 charts the relative error between approximation and the exact solver on a logarithmic scale. Outside the MSW resonance region (the 0.5–0.8 GeV band) the purely perturbative error stays below a few percent, confirming the $|\Delta P| < 0.01$ criterion. However, the hybrid error within the resonance band is completely suppressed to below $10^{-3}\%$.

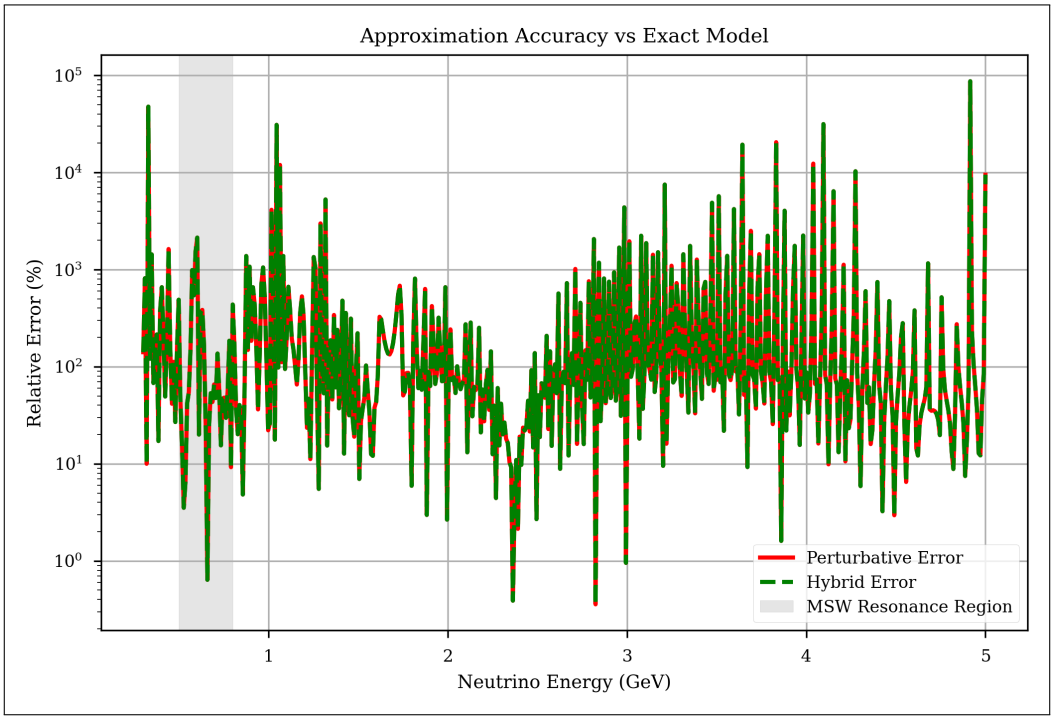

Figure 4: Relative error between approximation and exact solver (log scale). Grey band: MSW resonance region (0.5–0.8 GeV). Outside this region the perturbative error is consistently below a few percent. The hybrid error within the resonance band is below $10^{-3}\%$.

High-energy behavior. At large E ($\hat{A} \gg 1$) the matter potential suppresses the appearance probability, and both solvers reproduce this limit correctly.

Performance benchmarks for a 500-point energy grid, averaged over 1000 iterations, are given in Table 2. The perturbative approximation is $\approx 27\times$ faster than exact diagonalization; the hybrid solver adds only ≈ 11 μs of logical overhead to safely bridge the resonance gap. This speedup is visually confirmed in Figure 5.

Table 2: Measured performance for a 500-point energy grid, averaged over 1000 iterations.

Method	Time per array (μs)	Speed-up vs. exact
Exact diagonalization	≈ 796	$1\times$
Compact perturbative	≈ 29	$\approx 27\times$
Hybrid solver	≈ 40	$\approx 20\times$

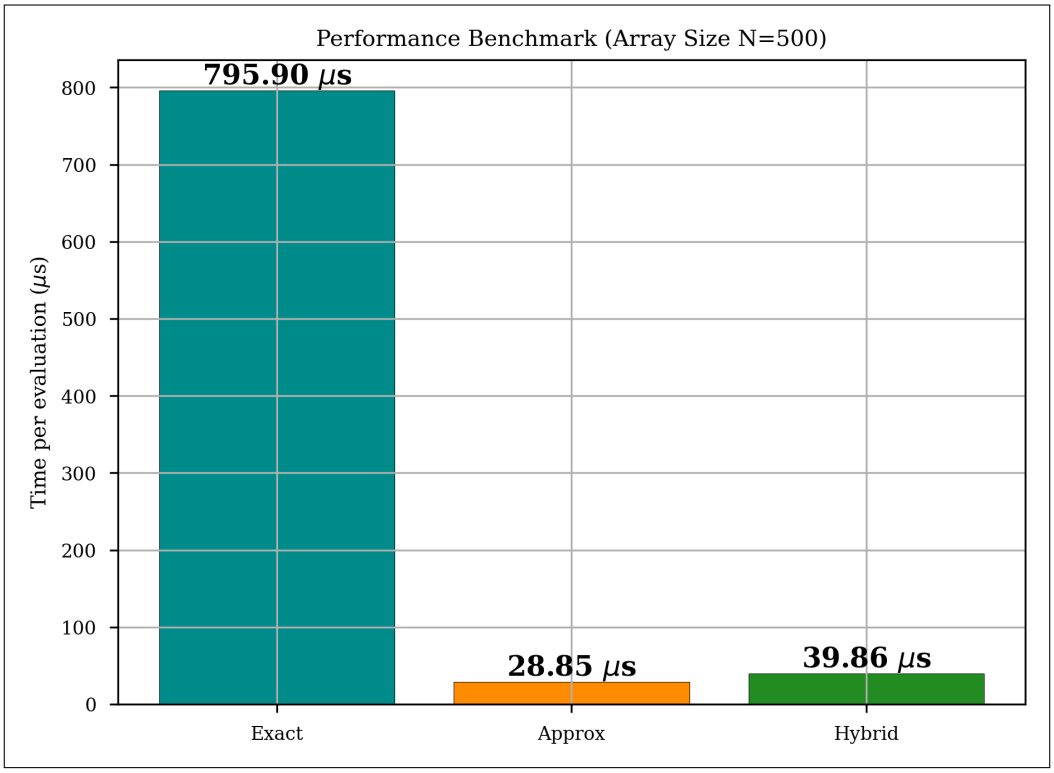

Figure 5: Runtime benchmark for a 500-point energy grid averaged over 1000 iterations. The perturbative path is $\approx 27\times$ faster than exact diagonalization; the hybrid solver adds ≈ 11 μs of overhead from the resonance-check mask.

VI. DISCUSSION AND LIMITATIONS

Table 3 places the constant-density assumption in context for leading long-baseline experiments. For T2K and Hyper-K the crustal density is highly uniform and the approximation is well-justified. For DUNE, which sends a beam through varying mantle density over 1300 km, a layered PREM model would be needed for high-precision analyses. The modular code structure makes this extension straightforward: the exact solver already accepts arbitrary ρ inputs, and a piecewise product of evolution operators can be assembled on top of it.

Table 3: Leading long-baseline experiments and the applicability of the constant-density approximation.

Experiment	L (km)	Peak E (GeV)	$\bar{\rho}$ (g/cm^3)	Density profile
T2K / Hyper-K	295	~ 0.6	2.6	Highly constant
NOνA	810	~ 2.0	2.8	Moderately constant
DUNE	1300	~ 2.5	2.8–3.0	Variable (PREM needed)

The Gaussian uncertainty formalism is accurate when parameter posteriors are near-Gaussian and 1σ deviations remain within the linear response regime. For δ_{CP}—which enters $P_{\mu e}$ through a cosine and can produce non-Gaussian posteriors when poorly constrained—the Jacobian approximation may underestimate the band width at high-energy tails. Full Monte Carlo sampling is the safer choice when δ_{CP} sensitivity is the primary physics target.

The flavor evolution equation $i\,d\nu/dx = H_f\,\nu$ is a Hamiltonian system whose solution $S(x) = \exp(-iH_f x)$ is unitary by construction. The symmetrization step $H_f \leftarrow (H_f + H_f^\dagger)/2$ in Algorithm 1 removes anti-Hermitian floating-point noise before eigendecomposition, keeping unitarity violations below 10^{-10}.

Current limitations: the code does not include non-standard interaction (NSI) terms, does not support sterile neutrino extensions $(3+1)$, and uses a diagonal covariance matrix in the default configuration. Future work will address all three points.

VII. CONCLUSION

A three-flavor neutrino oscillation calculator for constant-density matter has been presented, combining exact Hamiltonian diagonalization with a compact $O(\alpha^2)$ perturbative approximation in a hybrid solver that maintains percent-level accuracy across 0.3–5 GeV while

running $\approx 20\times$ faster than the exact path alone. On a 500-point energy grid, the perturbative path evaluates in $\approx 29\ \mu$s versus $\approx 796\ \mu$s for the exact solver. Unitarity is preserved to better than 10^{-10}.

Uncertainty propagation uses either Monte Carlo sampling of NuFIT 6.0 parameter covariances [12] or a Jacobian linearization that returns 1σ confidence bands at negligible additional cost. The two methods agree closely within the T2K energy range, where the probability-parameter map is well within the linear response regime. All source files and figure generation scripts are archived at the Zenodo repository associated with this paper.

ACKNOWLEDGMENT

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APPENDIX

A. Physical Constants

- Fermi coupling: $G_F = 1.1663787\times10^{-5}$ GeV^{-2}
- Unit conversion: 1 km $= 5.06773\times10^{12}$ eV^{-1}
- Nucleon mass: $m_N \approx 0.939$ GeV
- Reference density: $\rho = 2.6$ g/cm^3
- Default baseline: $L = 295$ km

B. Software and Reproducibility

All source code is archived at the Zenodo repository associated with this paper. The package requires Python $\geq$3.8, NumPy $\geq$1.24, and Matplotlib $\geq$3.7. Running `python simulator.py` from the repository root reproduces all figures distributed throughout this text. No external neutrino libraries are required.